Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/12/2016
PI Name:	Motil, Brian J. Ph.D.		
Project Title:	Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brian.j.motil@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	216-433-6617
Organization Name:	NASA Glenn Research Center		
PI Address 1:	21000 Brookpark Road		
PI Address 2:	GRC-LTZ0		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/12/2010	**End Date:**	10/11/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balakotaiah , Vemuri (University of Houston (grant NNX10AL37G))		
Grant/Contract No.:	Internal Project ; CoI grant NNX10AL37G		
Performance Goal No.:			
Performance Goal Text:			
	A basic understanding of two-phase flows through porous media is of interest in many chemical and biological processing systems as well as many geophysical applications. NASA technologies include adiabatic two-phase flows or two-phase flows with heat transfer that are relevant to life support systems, fuel cells, in-situ resource utilization, heat pipes, materials processing, production of pharmaceutical grade water, and transport of nutrients in soils. A very important unit operation that involves gas-liquid flow through porous media is the fixed Packed Bed Reactor (PBR). Examples of current space systems in which a fixed PBR is used include the Volatile Removal Assembly (VRA), the Integrated Advanced Water Recovery System (AWRS), and IntraVenous Water GENeration system (IVGEN). NASA is currently operating systems with packed bed reactors for ground and flight-based testing with an incomplete understanding of how the reduced gravity environment affects the performance and reliability of these reactors. This is		

Task Description:	especially critical when gas-liquid flows are involved. The expected outcome of this research effort is to develop a set of guidelines and tools to enable engineers to reliably design and operate packed bed reactors for microgravity as well as the lunar and Martian environments. Existing approaches for predicting the flow regimes and pressure drop for gas-liquid flow through a PBR are valid only for 1 g, and are based on liquid (or gas) holdup within the bed. Initial microgravity tests conducted in the reduced gravity aircraft and drop tower clearly show that the liquid holdup is significantly influenced by gravity. Furthermore, most of these models have only been verified using the "trickle flow" regime which does not exist in microgravity. Basic hydrodynamic models developed specifically for reduced gravity have been partially validated within the constraints of aircraft and ground facilities. However, typical flow ranges used for real systems require up to several minutes to develop fully which is far beyond the time scales available for ground tests. In addition, instabilities and hysteresis effects that may exist require long testing durations to establish new operating conditions. Thus, testing on the International Space Station (ISS) is required for complete validation of the semi-empirical models. Flight experiments can provide critical hydrodynamic information for a project with a broader scope which includes reduced gravity aircraft and ground-based (1 g) experiments. The main objective of the overall project is to develop and validate macroscopic equations that can be used in partial and microgravity conditions to accurately predict flow pattern transitions; pressure drops; and mass transfer rates in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. The hydrodynamic investigations addressed in the Packed Bed Reactor Experiment (PBRE) focus on the transitions between flow regimes (e.g., bubbly-to-pulse flow transition) and the associated pressure gradients over the range of relevant operating regimes of the PBRs (e.g., liquid and gas flow rates, particle sizes, and fluid viscosities). These design tools provide a fundamental understanding of these phenomena in low gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	(Ed. note 1/12/2016: Compiled from reporting dated February 2014 for research performed April 2010 through February 2014. Publications in Bibliography section are from 2010-February 2014). The main objective of the Packed Bed Reactor Experiment (PBRE) flight project at University of Houston (UH), with Co-Investigator V. Balakotaiah, grant NNX10AL37G, was to develop and validate macroscopic equations that can be used in microgravity conditions to accurately predict flow pattern transitions; pressure drop; and mass transfer rates in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. (A): PBRE Experiment: During the grant period, the Co-I (V. Balakotaiah) and the post-doctoral fellow (Dr. Salgi) have worked with the “Packed-Bed Reactor Team” at NASA Glenn Research Center towards achieving the main objective of the PBRE flight experiment. Specifically, both have participated in the discussions of the PBRE team related to the flight experiment and in finalizing the Science Requirements Document (SRD). In addition, the Co-I and Dr. Salgi have worked on the following specific objectives at the University of Houston. (B) Modeling Studies: Developed and validated a two-phase flow model for predicting the impact of gravity on pressure drop, liquid holdup, and regime flow transitions in two-phase gas-liquid flows through randomly packed beds. (C): Aircraft Experiments: The Co-I and Dr. Salgi have also assisted the PBRE team with the aircraft two-phase flow experiments that were conducted in May of 2012 and have since been helping with the data analysis effort aimed at characterizing pulse properties under microgravity conditions. A manuscript based on this data is under preparation. (D) Experimental Mass Transfer Studies at UH: Developed and tested mass transfer probes for measuring the gas to liquid mass transfer in normal gravity two-phase gas-liquid flows through packed-beds. Refereed Journal Publications: The following refereed journal publications acknowledge the support of this specific grant. 1. P. Salgi and V. Balakotaiah, "Impact of Gravity on Bubble-to-Pulse Transition in Packed Beds", AIChE Journal, 60, pp. 778-793 (2014). 2. C. Meza and V. Balakotaiah, "Celerity-Amplitude Relations for Solitary Waves on Vertically Falling Films", Industrial and Engineering Chemistry Research, 50, 13258-13272 (2011) Manuscript in preparation: 3. P. Salgi, V. Balakotaiah, E. Rame and B. J. Motil, “Pulse Flow Properties in Randomly Packed Beds under Microgravity Conditions”, Int. J. Multiphase Flow, to be submitted. New grant number for Co-I Balakotaiah is NNX14AD28G.
Bibliography Type:	Description: (Last Updated: 04/26/2024)
Articles in Peer-reviewed Journals	Salgi P, Balakotaiah V. "Impact of gravity on the bubble-to-pulse transition in packed beds." AlChE Journal. 2014 Feb;60(2):778-93. http://dx.doi.org/10.1002/aic.14300 , Feb-2014
Articles in Peer-reviewed Journals	Meza CE, Balakotaiah V. "Celerity-amplitude relations for solitary waves on vertically falling films." Industrial and Engineering Chemistry Research. 2011 Dec 7; 50(23)13258-72. http://dx.doi.org/10.1021/ie2005803 , Dec-2011
Papers from Meeting Proceedings	Patton M, Bruzas A, Rame E, Motil B. "Development of Packed Bed Reactor ISS Flight Experiment." 50th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition, Nashville, TN, January 9-12, 2012. 50th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition, Nashville, TN, January 9-12, 2012. AIAA paper 2012-0589. http://dx.doi.org/10.2514/6.2012-589 , Jan-2012